

MID

Course: Data Structures (CS101)

Set: A | Time: 2 Hours | Max Marks: 60 | Date: 15-03-2026

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

GENERAL INSTRUCTIONS:
attend all the questions

SECTION A: MCQs

1. Architects require a diverse skill set beyond purely technical abilities to successfully engage with stakeholders and manage system development. Which of the following skills is explicitly mentioned as essential for architects? **[5 Marks]**

- a) Advanced coding proficiency and debugging techniques.
- b) Diplomacy, negotiation, and communication skills.
- c) Financial accounting and project budgeting expertise.
- d) Expertise in hardware design and manufacturing processes.

2. According to the text, which statement accurately reflects an implication of the definition of software architecture? **[5 Marks]**

- a) Every computing system with software necessarily has a known and documented software architecture.
- b) Architecture solely defines the internal implementation details of software elements, suppressing external interactions.
- c) A system can only have one architectural structure that claims to be its complete architecture.
- d) An architecture defines software elements, how they relate, and can exist independently of its description or specification.

3. In the context of "Component-and-Connector Views," what is the term used for elements that have some runtime presence, such as processes, objects, clients, servers, and data stores? **[5 Marks]**

- a) Modules
- b) Views
- c) Components
- d) Connectors

4. The text categorizes various quality attributes. Which of the following is explicitly listed under "Business Quality Attributes"? **[5 Marks]**

- a) Modifiability
- b) Security
- c) Time to Market
- d) Usability

5. What is the primary focus of a "Quality View" as opposed to structural views like Module, C&C, or Allocation views? **[5 Marks]**
- a) It primarily shows the hierarchical decomposition of modules into smaller submodules.
 - b) It is tailored for specific stakeholders or concerns by extracting relevant pieces of structural views.
 - c) It exclusively maps software units to elements of an execution environment, like hardware.
 - d) It describes how larger modules are decomposed into smaller ones recursively.
6. During the "Reconstructing Software Architecture" process, what is the purpose of the "View fusion" activity? **[5 Marks]**
- a) To extract information from various source artifacts like source code or documentation.
 - b) To convert extracted information into a standard tuple-based or SQL-based database format.
 - c) To combine information in the database to produce a coherent view of the architecture.
 - d) To generate an architecture representation by building abstractions and various data representations.

SECTION B: SHORT ANSWERS

7. The text defines software architecture and discusses its implications. Briefly explain the core components that comprise the software architecture of a program or computing system, and then state one implication of this definition regarding abstraction. **[6 Marks]**
8. Differentiate between an Architectural Pattern, a Reference Model, and a Reference Architecture by providing the definition for each and explaining how they relate to, but are distinct from, an actual architecture. **[6 Marks]**
9. According to the text, what is the fundamental purpose of "Module Views" within architectural structures, and what are two properties that help to guide their implementation or analysis? **[6 Marks]**
10. The text highlights that architecture documentation serves multiple critical uses. Explain any three of these uses, as described in the source text. **[6 Marks]**
11. The Attribute-Driven Design (ADD) approach is described as a method for defining software architecture. Identify the key inputs required for ADD and specify what the output of this recursive decomposition process is. **[6 Marks]**